

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

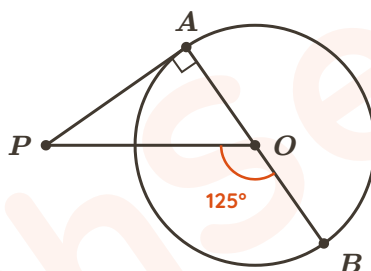
## Chapter 10: Circles

Previous year board questions, topic-wise: 27 questions

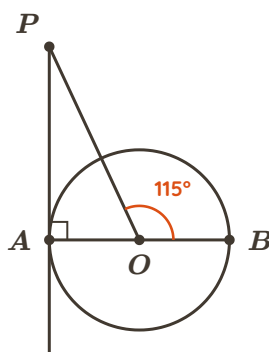
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

### 1 Theorem 1: tangent and radius

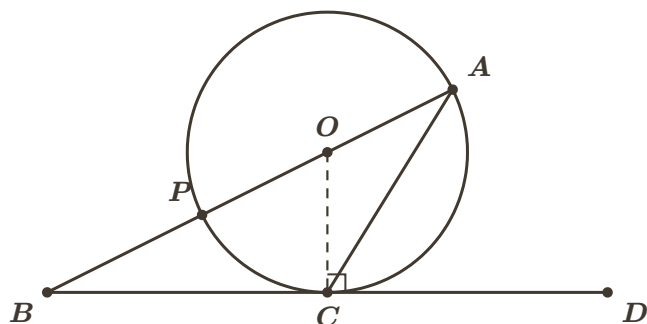
1.  $PA$  is a tangent from an external point  $P$  to a circle with centre  $O$ , touching it at  $A$ .  $AOB$  is a diameter and  $\angle POB = 125^\circ$ . Find  $\angle APO$ . (a)  $25^\circ$  (b)  $65^\circ$  (c)  $90^\circ$  (d)  $35^\circ$  CBSE 2026, 1 mark



2. Two concentric circles have radii 5 cm and 4 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. CBSE 2026, 2 marks
3. Tangents are drawn to a circle at the two ends of a diameter. How are the two tangents related? CBSE 2025, 1 mark
4.  $AOB$  is a diameter of a circle with centre  $O$ , and  $PA$  is the tangent at  $A$ . If  $\angle POB = 115^\circ$ , find  $\angle APO$ . CBSE 2025, 1 mark



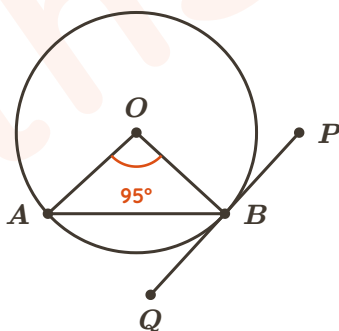
5.  $O$  is the centre of a circle and  $BCD$  is the tangent to it at  $C$ . The line from  $B$  through  $O$  meets the circle at  $P$  and  $A$ . Prove that  $\angle BAC + \angle ACD = 90^\circ$ . **CBSE 2025, 2 marks**



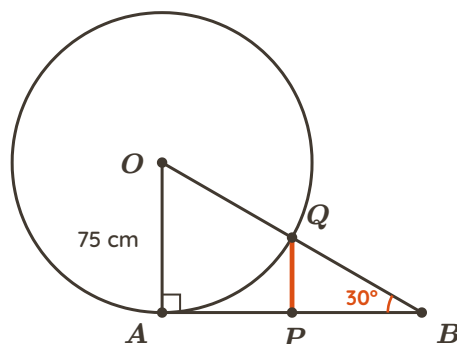
6. Assertion (A): the tangents at the two ends of a diameter of a circle are parallel. Reason (R): the diameter is the longest chord of a circle. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true.

**CBSE 2024, 1 mark**

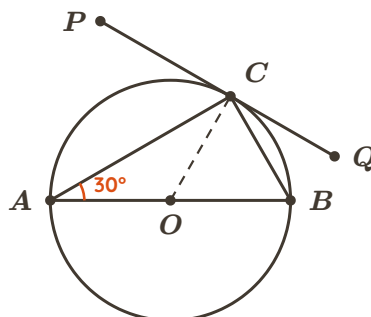
7.  $PQ$  is the tangent at  $B$  to a circle with centre  $O$ , and  $AB$  is a chord with  $\angle AOB = 95^\circ$ .  $Q$  lies on the same side of  $OB$  as  $A$ . Find  $\angle ABQ$ . (a)  $47.5^\circ$  (b)  $42.5^\circ$  (c)  $85^\circ$  (d)  $95^\circ$  **CBSE 2023, 1 mark**



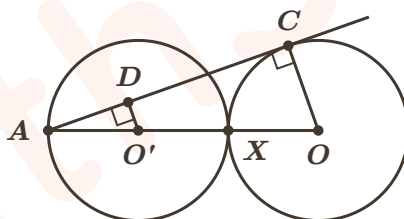
8. A discus leaves the thrower's circular path along a tangent.  $AB$  is such a tangent, touching a circle of radius 75 cm and centre  $O$  at  $A$ , with  $\angle ABO = 30^\circ$ .  $OB$  cuts the circle at  $Q$ , and  $P$  is the point on  $AB$  with  $PQ \parallel OA$ . (i) Find  $AB$ . (ii) Find  $OB$ . (iii) Find  $AP$ , OR find  $PQ$ . **CBSE 2023, 4 marks**



9. Two concentric circles have radii 4 cm and 3 cm. Find the length of the chord of the larger circle that touches the smaller circle. **CBSE 2022 Term 2, 2 marks**
10.  $PQ$  is the tangent at  $C$  to a circle with centre  $O$ , and  $AB$  is a diameter with  $\angle CAB = 30^\circ$ .  $P$  lies on the same side of  $OC$  as  $A$ . Find  $\angle PCA$ . **CBSE 2016, 1 mark**

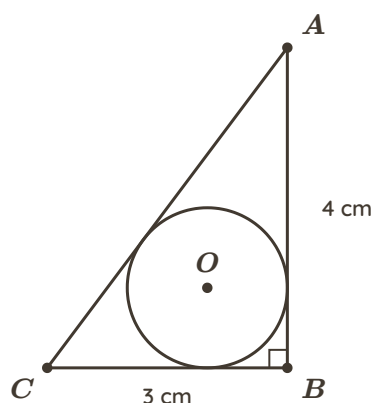


11. Two equal circles with centres  $O$  and  $O'$  touch each other at  $X$ . The line  $OO'$  produced meets the circle with centre  $O'$  again at  $A$ .  $AC$  is a tangent from  $A$  to the circle with centre  $O$ , touching it at  $C$ , and  $O'D \perp AC$ . Find  $\frac{DO'}{CO}$ . **CBSE 2016, 4 marks**

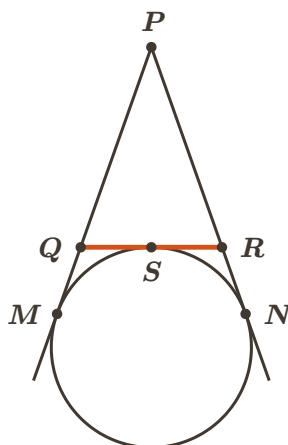


## 2 Theorem 2: tangents from an external point

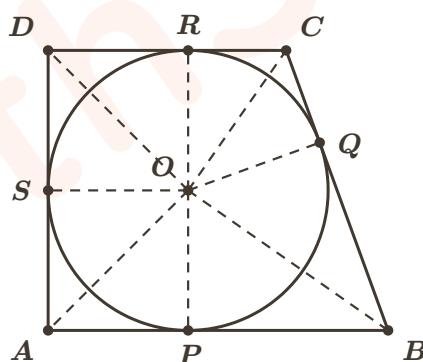
12.  $TP$  and  $TQ$  are tangents from an external point  $T$  to a circle with centre  $O$ , and  $\angle POQ = 120^\circ$ . Find  $\angle PTQ$ . (a)  $60^\circ$  (b)  $70^\circ$  (c)  $80^\circ$  (d)  $90^\circ$  **CBSE 2026, 1 mark**
13.  $\triangle ABC$  is right-angled at  $B$  with  $AB = 4$  cm and  $BC = 3$  cm. Find the radius of the circle inscribed in the triangle. **CBSE 2026, 3 marks**



14. A circle touches the side  $QR$  of  $\triangle PQR$  at  $S$ , and touches  $PQ$  and  $PR$  produced at  $M$  and  $N$ .  
Prove that  $PM = \frac{1}{2}(PQ + QR + PR)$ . **CBSE 2026, 3 marks**

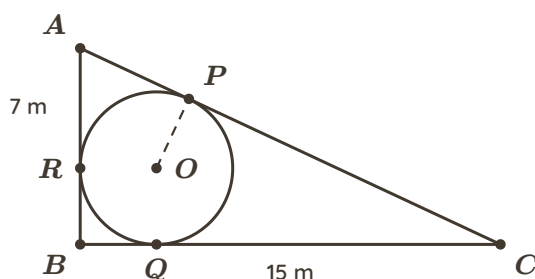


15. A point  $P$  is 26 m from the centre of a circle. Each tangent from  $P$  to the circle is 10 m long. Find the radius of the circle. **CBSE 2025, 2 marks**
16. A quadrilateral is drawn around a circle so that all four sides touch it. Prove that each pair of opposite sides subtends supplementary angles at the centre. **CBSE 2025, 2 marks**

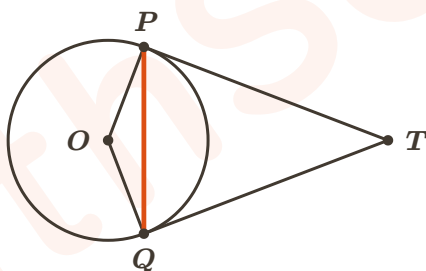


17.  $AB$  is a diameter of a circle with centre  $O$ . The tangents at  $A$  and  $B$  are met by a third tangent at  $Q$  and  $P$  respectively. Prove that  $\angle POQ = 90^\circ$ . **CBSE 2024, 3 marks**
18. A circle of radius 8 cm with centre  $O$  touches the sides  $AD$ ,  $DC$ ,  $CB$  and  $BA$  of a quadrilateral  $ABCD$  at  $P$ ,  $Q$ ,  $R$  and  $S$ .  $AD \perp DC$ ,  $BC = 30$  cm and  $BS = 24$  cm. Find  $DC$ . **CBSE 2024, 3 marks**

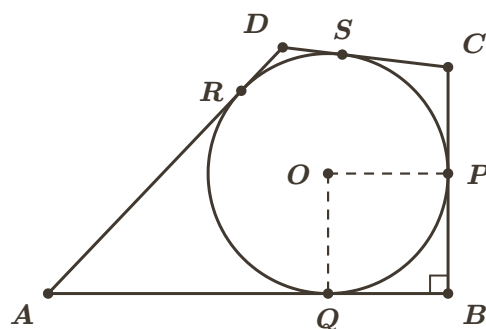
19. A triangular backyard  $ABC$  has a right angle at  $B$ , with  $AB = 7$  m and  $BC = 15$  m. A circular pit inside it touches  $AC$ ,  $BC$  and  $AB$  at  $P$ ,  $Q$  and  $R$ , and  $AP = x$  m. (i) Write  $AR$  in terms of  $x$ . (ii) What kind of quadrilateral is  $BQOR$ ? (iii) Write  $PC$  in terms of  $x$  and find  $x$ , OR find  $x$  and then the radius  $r$  of the pit. **CBSE 2024, 4 marks, case study**



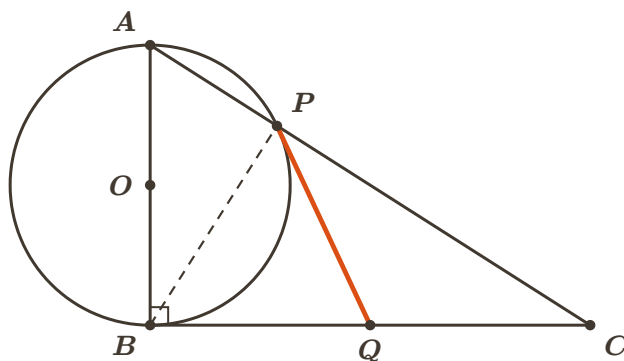
20.  $TP$  and  $TQ$  are tangents from an external point  $T$  to a circle with centre  $O$ . Prove that  $\angle PTQ = 2\angle OPQ$ . **CBSE 2023, 3 marks**



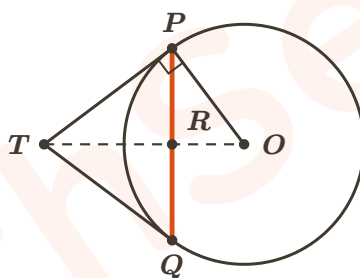
21. A circle with centre  $O$  is inscribed in a quadrilateral  $ABCD$ , touching  $AB$ ,  $BC$ ,  $CD$  and  $DA$  at  $Q$ ,  $P$ ,  $S$  and  $R$ .  $\angle B = 90^\circ$ ,  $AD = 17$  cm,  $AB = 20$  cm and  $DS = 3$  cm. Find the radius of the circle. **CBSE 2023, 3 marks**



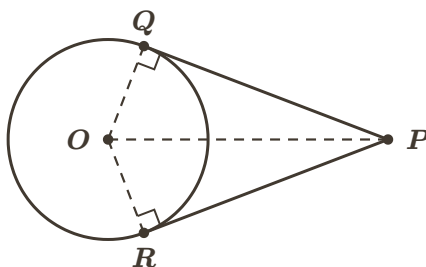
22.  $\triangle ABC$  is right-angled at  $B$ . A circle drawn on  $AB$  as diameter cuts  $AC$  at  $P$ . The tangent to the circle at  $P$  meets  $BC$  at  $Q$ . Prove that  $Q$  is the mid-point of  $BC$ . CBSE 2022 Term 2, 4 marks



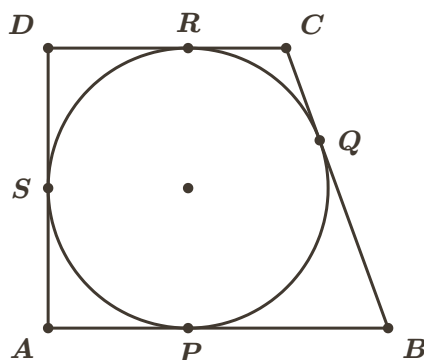
23.  $PQ$  is a chord of length 8 cm in a circle of radius 5 cm with centre  $O$ . The tangents at  $P$  and  $Q$  meet at  $T$ . Find  $TP$ . CBSE 2019, 3 marks



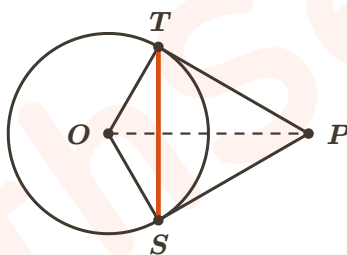
24. Prove that the two tangents drawn to a circle from a point outside it are equal in length. CBSE 2018, 3 marks



25. A quadrilateral  $ABCD$  is drawn around a circle, with  $AB$ ,  $BC$ ,  $CD$  and  $DA$  touching the circle at  $P$ ,  $Q$ ,  $R$  and  $S$ . Prove that  $AB + CD = BC + DA$ . **CBSE 2016, 2 marks**



26.  $PT$  and  $PS$  are tangents from an external point  $P$  to a circle with centre  $O$  and radius  $r$ . If  $OP = 2r$ , show that  $\angle OTS = \angle OST = 30^\circ$ . **CBSE 2016, 2 marks**



27. Prove that the tangents drawn to a circle from an external point are equal in length. **CBSE 2016, 4 marks**

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